

DEERING, AK, USA

WMO: 701333

Lat: 66.069N

Lon: 162.764W

Elev: 6

StdP: 101.25

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 26643

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.2	-34.5	-36.3	0.1	-33.2	-35.1	0.1	-32.0	16.0	-9.8	14.6	-11.2	2.2	220	0.908

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.1	(c) 20.3	(d) 14.8	(e) 18.5	(f) 13.9	(g) 17.0	(h) 13.0	(i) 15.5	(j) 19.2	(k) 14.4	(l) 17.7	(m) 13.4	(n) 16.4	(o) 4.8	(p) 40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	9.7	17.5	12.5	9.0	16.3	11.5	8.5	15.3	43.5	19.2	40.3	17.7	37.6	16.4	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.7	12.0	10.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-38.9	24.2	2.4	3.1	-40.6	26.5	-42.1	28.3	-43.4	30.0	-45.2	32.3
			-35.4	16.3	1.0	1.3	-36.1	17.3	-36.7	18.0	-37.3	18.8	-38.0	19.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.4	-18.8	-16.4	-17.1	-9.7	0.5	8.3	11.9	10.1	5.8	-1.8	-10.5	-15.8		
	DBStd	12.94	9.95	9.83	7.81	6.91	5.30	3.77	2.81	2.99	3.46	5.06	6.97	8.84		
	HDD10.0	5396	893	738	841	592	295	76	14	37	131	366	614	799		
	HDD18.3	8295	1152	972	1099	842	553	302	199	256	376	624	864	1057		
	CDD10.0	143	0	0	0	0	1	24	73	40	5	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	9	0	0	0	0	0	1	7	2	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	4.7	5.1	5.8	5.1	4.9	4.4	4.1	4.1	4.3	4.6	4.6	4.6	5.1		
(15) Precipitation	PrecAvg	309	17	22	13	15	12	16	41	59	40	31	22	21		
	PrecMax	437	36	109	55	47	37	45	95	139	73	86	59	60		
	PrecMin	212	1	1	2	2	2	1	11	3	10	5	1	8		
	PrecStd	60	11	23	11	11	9	11	21	34	19	18	16	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	7.4	2.9	6.3	16.0	22.2	23.8	21.2	16.1	9.5	3.8	3.3		
		MCWB	1.7	3.4	1.7	3.7	9.6	14.3	16.5	14.7	11.4	6.8	2.1	1.7		
	2%	DB	1.1	2.1	0.5	3.2	12.4	19.1	20.8	18.3	13.9	7.5	2.3	1.0		
		MCWB	-0.3	0.2	-0.6	1.4	8.1	13.3	15.4	14.3	10.8	5.8	1.0	-0.3		
	5%	DB	-1.6	0.2	-1.6	2.0	9.9	17.1	19.3	16.9	12.6	6.2	1.1	-0.9		
		MCWB	-2.7	-1.2	-2.8	0.6	6.7	12.2	14.7	13.6	9.9	4.7	-0.1	-2.2		
	10%	DB	-4.0	-2.5	-3.8	0.5	7.8	15.1	17.6	15.4	11.3	4.7	-0.9	-3.3		
		MCWB	-5.1	-3.8	-4.8	-0.7	5.1	11.0	13.6	12.7	9.1	3.5	-1.8	-4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	3.8	1.9	3.8	9.9	15.7	17.4	15.8	12.3	7.4	2.2	1.8		
		MCDB	3.5	7.9	3.1	6.3	14.8	19.5	21.5	19.1	14.4	9.1	3.4	3.0		
	2%	WB	0.0	0.3	-0.5	1.7	8.4	13.8	16.2	14.9	11.2	5.9	1.0	-0.2		
		MCDB	1.2	1.9	0.5	3.2	12.3	18.4	20.3	17.6	13.3	7.2	2.0	0.8		
	5%	WB	-2.4	-2.0	-3.3	0.6	6.8	12.5	15.0	14.0	10.2	4.8	-0.1	-2.2		
		MCDB	-1.3	-0.7	-2.2	1.9	9.7	16.5	18.8	16.4	12.4	6.1	0.9	-1.1		
	10%	WB	-4.8	-4.5	-6.2	-0.6	5.2	11.1	13.9	13.0	9.2	3.5	-1.6	-4.4		
		MCDB	-3.8	-3.3	-5.4	0.5	7.5	14.7	17.2	15.3	11.1	4.7	-0.9	-3.4		
(35) Mean Daily Temperature Range	MDBR		7.4	8.0	8.0	7.3	7.1	8.8	8.1	7.5	7.0	6.0	7.4	7.6		
	5% DB	MCDBR	8.3	8.6	8.0	7.5	11.0	12.2	12.4	9.8	8.5	5.5	6.0	7.6		
		MCWBR	7.6	7.8	8.5	6.3	7.2	7.0	7.3	6.4	5.8	4.4	5.4	6.8		
	5% WB	MCDBR	8.2	9.0	9.0	7.1	10.7	11.2	11.6	8.9	8.2	5.2	5.4	7.4		
		MCWBR	7.5	7.9	8.5	6.1	7.1	6.7	7.2	6.1	5.9	4.3	5.0	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.175	0.242	0.262	0.328	0.353	0.346	0.359	0.373	0.305	0.262	N/A	0.172		
	taud		1.900	2.132	2.218	2.062	2.136	2.362	2.419	2.391	2.531	2.356	N/A	1.918		
	Ebn at Noon		270	630	814	821	839	862	838	780	777	618	N/A	40		
	Edh at Noon		18	53	80	123	127	105	97	90	62	42	N/A	4		
(44) All-Sky Solar Radiation	RadAvg		0.10	0.69	2.47	4.56	5.69	6.08	4.73	3.46	2.15	0.78	0.17	0.01		
	RadStd		0.01	0.08	0.20	0.28	0.57	0.45	0.39	0.41	0.21	0.09	0.02	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+1.43	N/A	N/A	+1.00	+0.82	N/A	-475	-510	+65	+1

Nomenclature: See separate page